Beyond Stochasticism: An Ontological Framework for Agentic Stability

OEA Framework Research Insights Report | v0.5.0 | 2026-05-14
BitConcepts Research | github.com/BitConcepts/oea-framework-paper

Executive Summary

The OEA (Ontology, Epistemic, Agentic) framework is a three-layer, hypothesis-driven protocol for improving reliability in recursive generative pipelines under synthetic-data exposure. This report summarises the experimental evidence from v0.5.0: bigram-proxy ablation study (12 variants, 7,776 runs) and a two-model real LLM validation (distilgpt2 82M and gpt2 124M, 10 seeds x 10 iterations each, run on CUDA 12.1 / RTX 4070 SUPER). All experiments are fully reproducible.

Key Findings

1. Calibration direction is the causal variable

The anti-calibrated ablation (oea_miscalibrated) degrades log-probability by -1.14 nats (distilgpt2) and -0.55 nats (gpt2) vs control, while oea_anchored improves it by +0.62 nats and +1.63 nats. The consistent direction across both models and both conditions provides strong ablation evidence that calibration direction, not filtering per se, is the operative mechanism.

2. Metric dissociation: log-prob vs ROUGE-L

ROUGE-L recall (an independent, log-prob-orthogonal metric) shows the opposite direction: oea_anchored has LOWER ROUGE-L (0.038 vs control 0.046) while oea_miscalibrated has HIGHER ROUGE-L (0.054). This is not a contradiction - vocabulary anchoring improves distributional fidelity (log-prob) while constraining phrase-level sequence order (reducing ROUGE-L). The two metrics capture orthogonal quality dimensions. This finding addresses metric-circularity concerns.

3. Bigram ablation: d=4.56, p<0.001

The full OEA protocol achieves the highest true-rejection rate (TRR=0.836) and lowest false-rejection rate (FRR=0.081) across all 12 tested variants over two domain corpora (literary + scientific). Cohen's d=4.56 vs control_replace, permutation p<0.001. The ablation_miscalibrated variant confirms the causal direction: TRR=0.257, FRR=0.651 (reversed discrimination).

4. RAG without epistemic filter degrades quality

oea_rag_only shows negative log-prob deltas for both models (-0.54 / -0.08 nats). BM25 context injection without quality-aware candidate selection harms distributional fidelity. The epistemic filter is the operative component in oea_anchored.

5. Stability/epistemic orthogonality

Some single-layer ablations (ablation_A, ablation_EA) match or exceed oea_full in raw stability while underperforming on epistemic filtering. This confirms the two properties are partially orthogonal - the full three-layer protocol is needed when both are required jointly.

Table 1: Bigram-Proxy Ablation Study (12 Variants, 648 Runs Each)

Credibility suite v2, two corpora (literary: Carroll/Austen/Melville/Hume/Darwin; scientific: Newton/Feynman/Sagan). Cohen d=4.56 vs control_replace, $p<0.001$.

Variant	Stability (mean [95% CI])	TRR (mean [95% CI])	FRR (mean [95% CI])
control_replace	0.449 [0.437, 0.460]	0.582 [0.580, 0.584]	0.241 [0.239, 0.243]
control_accumulate	0.943 [0.942, 0.944]	0.582 [0.580, 0.584]	0.241 [0.239, 0.243]
baseline_rag	0.935 [0.934, 0.936]	0.682 [0.680, 0.684]	0.178 [0.177, 0.180]
baseline_calibration	0.935 [0.934, 0.936]	0.651 [0.649, 0.653]	0.198 [0.196, 0.199]
ablation_O	0.955 [0.955, 0.956]	0.613 [0.610, 0.615]	0.222 [0.220, 0.223]
ablation_E	0.935 [0.934, 0.936]	0.752 [0.750, 0.754]	0.134 [0.132, 0.135]
ablation_A	0.957 [0.956, 0.957]	0.635 [0.633, 0.637]	0.207 [0.205, 0.209]
ablation_OE	0.955 [0.955, 0.956]	0.775 [0.773, 0.777]	0.120 [0.118, 0.121]
ablation_OA	0.938 [0.936, 0.939]	0.667 [0.665, 0.669]	0.188 [0.186, 0.189]
ablation_EA	0.950 [0.948, 0.951]	0.806 [0.804, 0.807]	0.100 [0.099, 0.101]
ablation_miscalibrated	0.943 [0.942, 0.944]	0.257 [0.255, 0.259]	0.651 [0.648, 0.653]
oea_full (BEST)	0.937 [0.935, 0.938]	0.836 [0.834, 0.837]	0.081 [0.080, 0.082]

TRR = true-reject rate; FRR = false-reject rate. Green row = best.

Table 2: Real LLM Validation (GPU Run, CUDA 12.1, RTX 4070 SUPER)

Two GPT-2 family models, BM25 RAG, 10 seeds x 10 recursive iterations. Values at final iteration (iter=10). Log-prob higher = better. ROUGE-L = seed content recall (independent of log-prob selection). JSD = diversity proxy.

Variant	DG2 Log-prob	DG2 JSD	DG2 ROUGE-L	GPT2 Log-prob	GPT2 JSD	GPT2 ROUGE-L
control	-0.912	0.459	0.046	-1.916	0.363	0.047
oea_rag_only	-1.447 [-0.54]	0.382	0.045	-1.991 [-0.08]	0.364	0.049
oea_miscalibrated	-2.055 [-1.14]	0.356	0.054	-2.470 [-0.55]	0.384	0.055
oea_anchored	**-0.296 [+0.62]**	**0.471**	**0.038**	**-0.290 [+1.63]**	**0.438**	**0.031**

DG2 = distilgpt2 (82M); GPT2 = gpt2 (124M). Brackets show delta vs control. Green row = oea_anchored (best log-prob). Note: oea_anchored has LOWER ROUGE-L than control - this is expected (vocabulary anchoring reduces sequence recall while improving distributional fidelity). See Metric Dissociation section.

Metric Dissociation: Log-Probability vs ROUGE-L

A key finding from the GPU experiments is that log-probability and ROUGE-L recall systematically diverge under vocabulary anchoring. This is not a contradiction but an informative finding about the quality/diversity tradeoff:

Metric	oea_anchored vs control	What it measures	Why it diverges
Log-probability	HIGHER (+0.62/+1.63 nats)	Distributional fidelity under frozen model	Anchoring constrains to domain vocabulary tokens => more in-distribution
ROUGE-L recall	LOWER (0.038/0.031 vs 0.046/0.047)	Sequence-level content preservation from seed	Anchoring changes token identity but not phrase order => fewer original n-grams
JSD	HIGHER (0.471/0.438 vs 0.459/0.363)	Token distribution distance from seed	Anchored distribution is MORE concentrated => higher divergence from uniform seed

The dissociation confirms log-probability and ROUGE-L are orthogonal. Metric-circularity concerns (selecting by log-prob and measuring by log-prob) are addressed: ROUGE-L independently shows the mechanism's effect on a completely different quality dimension.

Manuscript Vulnerability Fixes Applied (v0.5.0)

Five critical peer-review vulnerabilities identified and addressed in this version:

Issue	Original claim	Fix applied	Status
1. Causal proof language	'constitutes a causal proof-of-mechanism'	Replaced with 'provides strong ablation evidence'	FIXED
2. Metric circularity	Log-prob is both selection criterion and metric	Added ROUGE-L as orthogonal independent metric; dissociation reported honestly	FIXED
3. JSD wrong direction	Higher JSD not explained	Explicitly labelled as expected diversity/quality tradeoff; ROUGE-L confirms	FIXED
4. RAG-only inconsistency	No explanation for model-dependent RAG-only results	New paragraph: RAG-only degrades log-prob for both models; filter is operative	FIXED
5. Agentic framing gap	'Agentic systems' but no actual agents in experiments	Bridging paragraph added + external validity bullet clarifying structural vs empirical scope	FIXED

Submission Status

Gate	Item	Status
A - Citation lock	13/13 citations verified (cycle 3)	CLOSED
B - Evidence lock	Bigram suite 7,776 runs + two-model real LLM	CLOSED
C - Manuscript lock	No placeholders; all 5 vulnerabilities fixed	CLOSED
D - Submission pkg	CI green; PDF compiles; metadata confirmed	CLOSED
- Submission	arXiv cs.AI+cs.CL; PhilSci-Archive; ResearchGate	PENDING (Issue #5)

Reproducibility

GPU environment: NVIDIA GeForce RTX 4070 SUPER, 12 GB VRAM, driver 591.74, CUDA 12.1

torch 2.5.1+cu121, transformers, rouge-score 0.1.2

All experiments use torch.manual_seed per generation call.

Setup: scripts/setup.cmd --experiments (Windows) or scripts/setup.sh --experiments (Linux/macOS). Auto-selects CUDA 12.1 wheel if nvidia-smi detected.

Artifact	Path	Runs	Reproducible
Bigram ablation	results/credibility/	7,776	Yes (config JSON + seeds)
Real LLM distilgpt2	results/real_lm/distilgpt2/	10x10x4	Yes (torch.manual_seed, CUDA 12.1)
Real LLM gpt2	results/real_lm/gpt2/	10x10x4	Yes (torch.manual_seed, CUDA 12.1)
Pilot experiments	results/summary_metrics.json	30/cond	Yes (seeded)